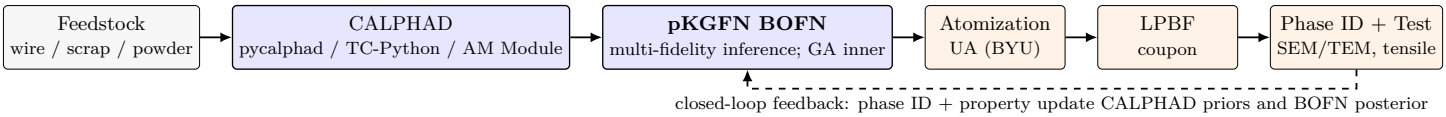
DSO Concept Outline — Metastable $L1_2/D0_{22}$ Strengthening Phases in Rapidly Solidified Dilute Al-MPEAs Inaccessible to Equilibrium CALPHAD

PI: Sterling Baird (Brigham Young University, Materials Science & Engineering). **Solicitation:** DARPA DSO Office-Wide BAA (1-page exec summary → invited full proposal); adjacent to METALS, SURGE. **Type:** seedling \$0.5–1M / 12–18 mo, with Phase 1 follow-on \$2–4M / 24 mo. **Keywords:** non-equilibrium phase selection, metastable $L1_2$, dilute Al-MPEA, BOFN, LPBF.



Fundamental technical barrier

The DARPA-hard claim is metallurgical, not logistical: *rapid-solidification thermal histories accessible via UA + LPBF stabilize coherent metastable $L1_2$-$Al_3(Zr,Ti,X)$ and $D0_{22}$-related ordered nanoprecipitates in dilute Al-Mn-Cr-Zr-Si-Ti space that current CALPHAD assessments rank as thermodynamically inaccessible at the bulk equilibrium level*, and these phases can deliver Al-Ce/Al-Ni-Sc-class elevated-temperature strength without REE/Sc additions. Equilibrium databases predict that Mn/Cr/Zr/Si/Fe additions cannot replicate the $L1_2/D0_{22}$ strengtheners that Sc/REE enable; the wager is that quenched-in supersaturation plus rapid LPBF re-melt cycles open a metastable solubility window in which these ordered phases nucleate at 10^4–10^6 K/s cooling rates, and that a closed BOFN-CALPHAD-experiment loop is the only tractable way to navigate that phase space without a combinatorial library. If the claim fails (no $L1_2/D0_{22}$-class precipitate observed by TEM/APT, or property delta <50% of comparator), no engineering wrap can rescue the program. Wire/scrap feedstock is downstream transition value, not the proposed DSO risk.

Idea

The FLAIME discovery loop — pKGFN BOFN over CALPHAD multi-fidelity nodes (`pycalphad`, TC-Python/TCAL10, AM Module), validated by UA and LPBF — applied to a constrained dilute Al-MPEA design space (Al + Mn/Cr/Zr/Ti/Si/Fe, ≤5 active solute elements). The objective function is dominated by (i) elevated-temperature UTS retention and (ii) TEM/APT-confirmed presence of the predicted ordered-precipitate family; printability window and feedstock convertibility are constraints, not design targets. **Hypothesis:** the loop, narrowed to a metastable-$L1_2/D0_{22}$ objective, can deliver one alloy candidate end-to-end (BOFN-CALPHAD design → UA powder → LPBF coupon → TEM/APT phase ID + elevated-temperature tensile) within a 12-month seedling, with the alloy occupying a phase region not predicted by current Al-MPEA databases.

Why DSO (fundamental risk, not integration risk)

Risk DSO is positioned to take: the metallurgical claim that rapid-solidification AM unlocks coherent metastable $L1_2/D0_{22}$ ordered nanoprecipitates in dilute Al-MPEA space unreachable by equilibrium thermodynamics. Component pieces (UA atomizer, LPBF baseline, BOFN) exist; their integration is engineering and is not the proposed DSO risk. **Seedling go/no-go (12 mo):** (i) one alloy candidate carried BOFN-CALPHAD design → UA powder → LPBF coupon → tensile + TEM/APT; (ii) independent test article printed at JHU APL using the resulting powder, characterized at BYU; (iii) TEM or atom-probe evidence of an $L1_2$- or $D0_{22}$-class precipitate predicted by the loop but absent from baseline CALPHAD assessment of the same composition; (iv) as-built UTS retention at 200 °C ≥ 85% of published Al-10Ce-3Ni and Al-2Sc-1Zr datasets at matched product form (LPBF coupon, no T6); (v) side-by-side public CALPHAD prediction showing the observed strengthening constituent should not exist at the measured cooling-rate history.

Preliminary results (FLAIME pipeline, BYU, Apr 2026)

AMAZEMET rePOWDER UA atomizer installed and operating at BYU; SEM/EDS on UA-modified powder (AlSi10Mg + Al wire + Si feed) showing measurable composition shift — the powder-modification pathway exists in working form. Elevated-temperature tensile data on a standard Al specimen; AlSi10Mg LPBF tensile baseline (printed at U. Utah Spear lab via no-cost MOU). Working pKGFN BOFN; Thermo-Calc AM Module simulations on candidate compositions; automated powder dispensing demonstrated.

Security posture and partnership

Funded: BYU (Baird, lead — BOFN, CALPHAD, UA, phase ID, tensile). **Funded transition partner:** JHU APL — independent print of the candidate alloy on an APL LPBF system, plus a transition pathway through APL's existing relationships with AFRL Manufacturing & Industrial Technologies Division and the Army DEVCOM Armaments Center; APL serves as the FFRDC-style bridge from seedling demonstration to a named DoD program office, not as the end customer. **Unfunded U.S. collaborator:** U. Utah Spear lab (no-cost LPBF baseline coupon access via existing MOU). Work is performed at BYU and APL on unclassified, fundamental-research-basis data (NSDD-189); no CUI/ITAR data are anticipated in the seedling. If a Phase 1 follow-on requires CUI handling, BYU has an established research-security office; APL operates at the required posture and would host any controlled data. **Adjacent venues acknowledged:** pure logistics-integration framings of this work are better placed at ONR (Naval Materials Division), Army DEVCOM, or DLA; the DSO submission is reserved for the fundamental metallurgical claim above.

Scope separation from pending DOE Genesis Mission Topic 3B and request

Task-level scope separation. *Genesis tasks (DOE-funded):* alloy-agnostic platform construction — hardware/software integration, scheduler, data fabric, autonomous orchestration, general platform validation across broad composition/process spaces. *DSO seedling tasks (this proposal):* (T1) hypothesis-bound CALPHAD-BOFN study scoped to the $L1_2/D0_{22}$ ordered-precipitate question in a defined dilute Al-MPEA window; (T2) powder + coupon production for the named candidate; (T3) TEM/APT phase ID and elevated-temperature tensile against published Al-Ce/Al-Ni-Sc benchmarks; (T4) independent APL print + characterization. Personnel labor, deliverables, data packages, and reporting artifacts are itemized separately; no scheduler or platform-development task appears in both. **Request:** a read on a 1-page exec summary before we invest in a full proposal.

Repository and full vision: https://github.com/vertical-cloud-lab.